AI/ML Task-1: Student Performance Data Explorer

1. Introduction

This project explores student performance data using Python and data analysis libraries.

2. Objectives

To load and clean student data, calculate descriptive statistics, and visualize relationships between student scores.

3. Dataset Description

The practice dataset contains student names and their math, reading, and writing scores.

4. Data Cleaning

Describe the missing-value counts and number of duplicate records found. Explain the cleaning operations performed.

5. Statistical Analysis

Include the mean, median, and mode of math scores, along with the descriptive statistics generated by Pandas.

6. Data Visualization

Include your histogram, scatter plot, box plot, and correlation heatmap. Explain what each graph shows.

7. Key Findings

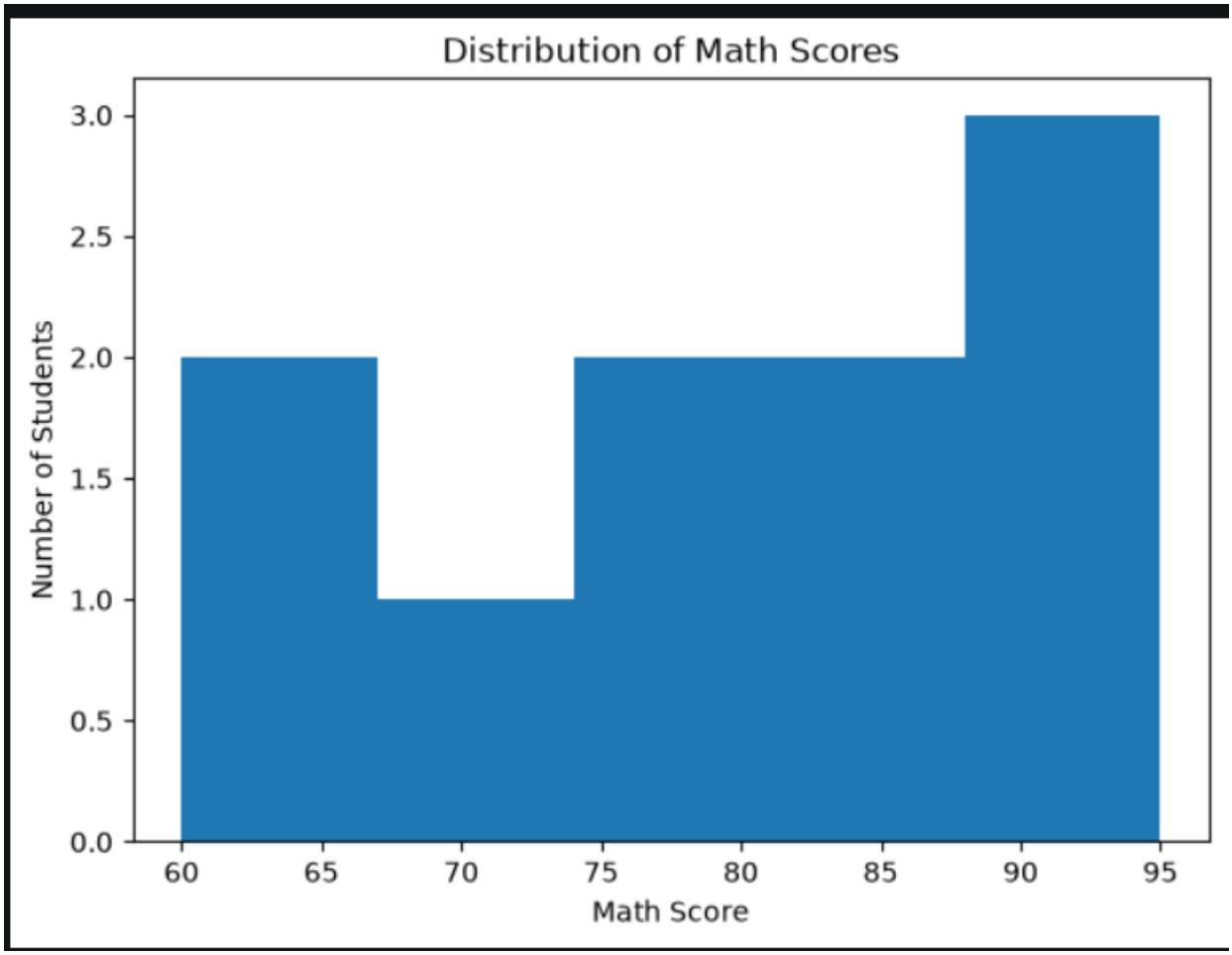
Write three to five observations based on the actual results from your notebook.

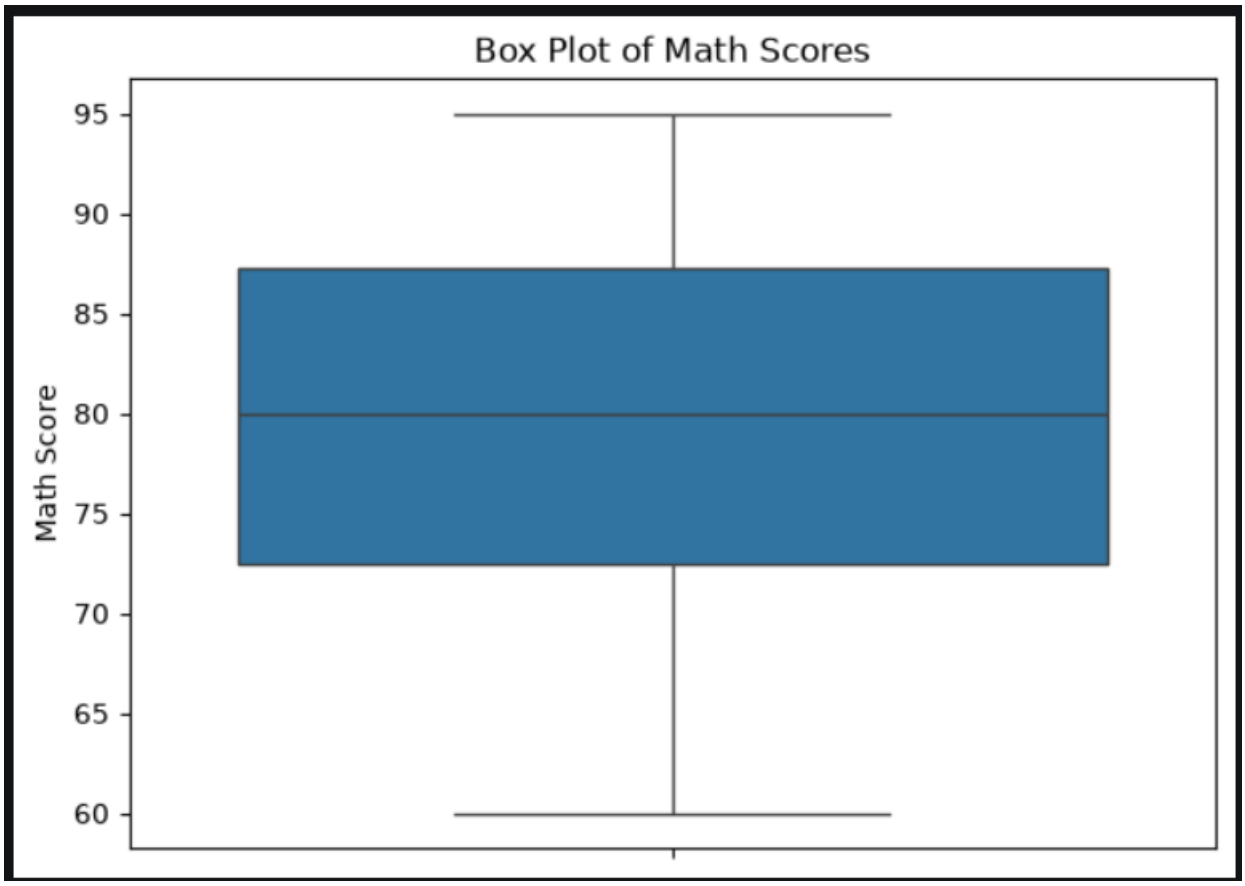
8. Conclusion

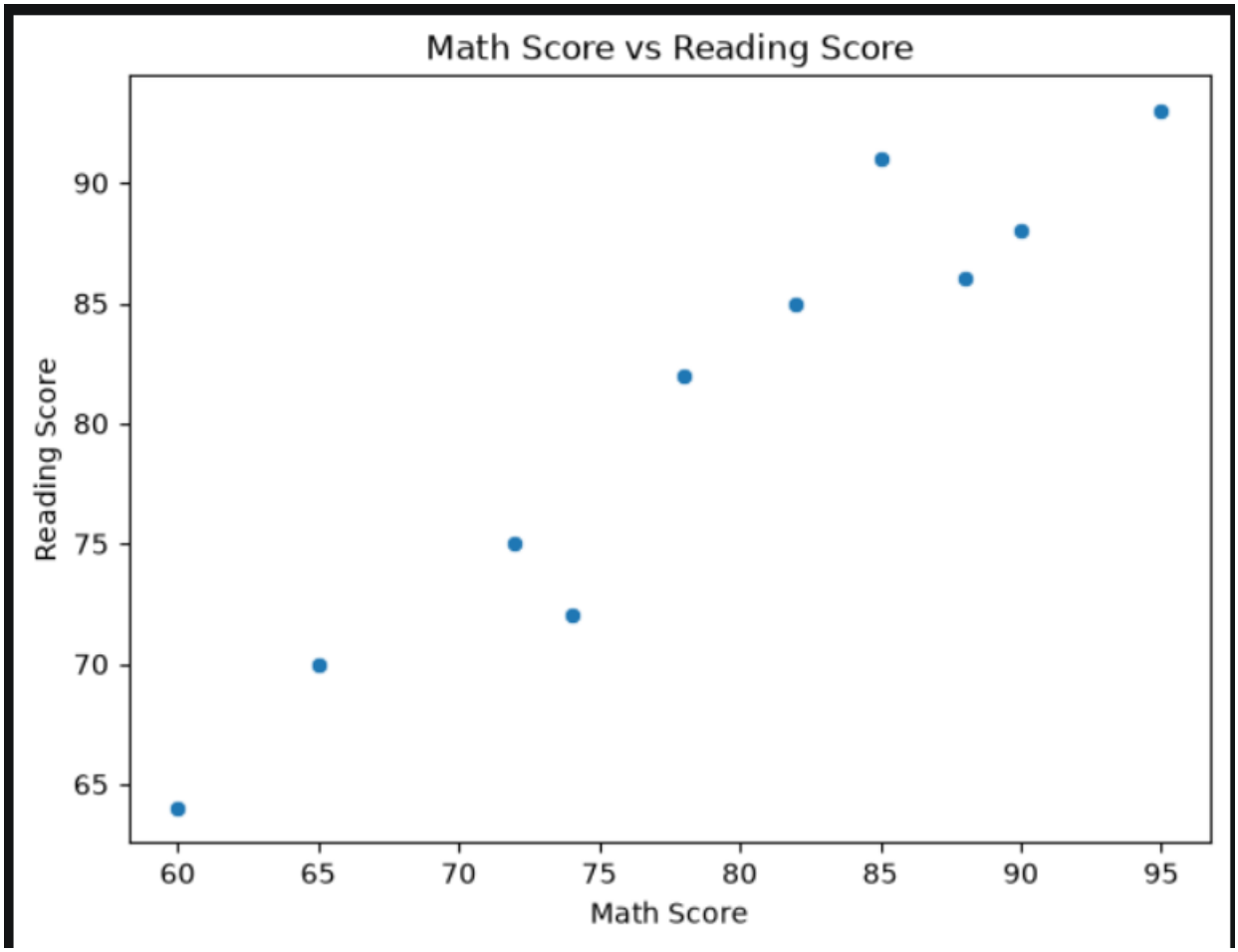
Summarize what you learned about loading, cleaning, analyzing, and visualizing data using Python.

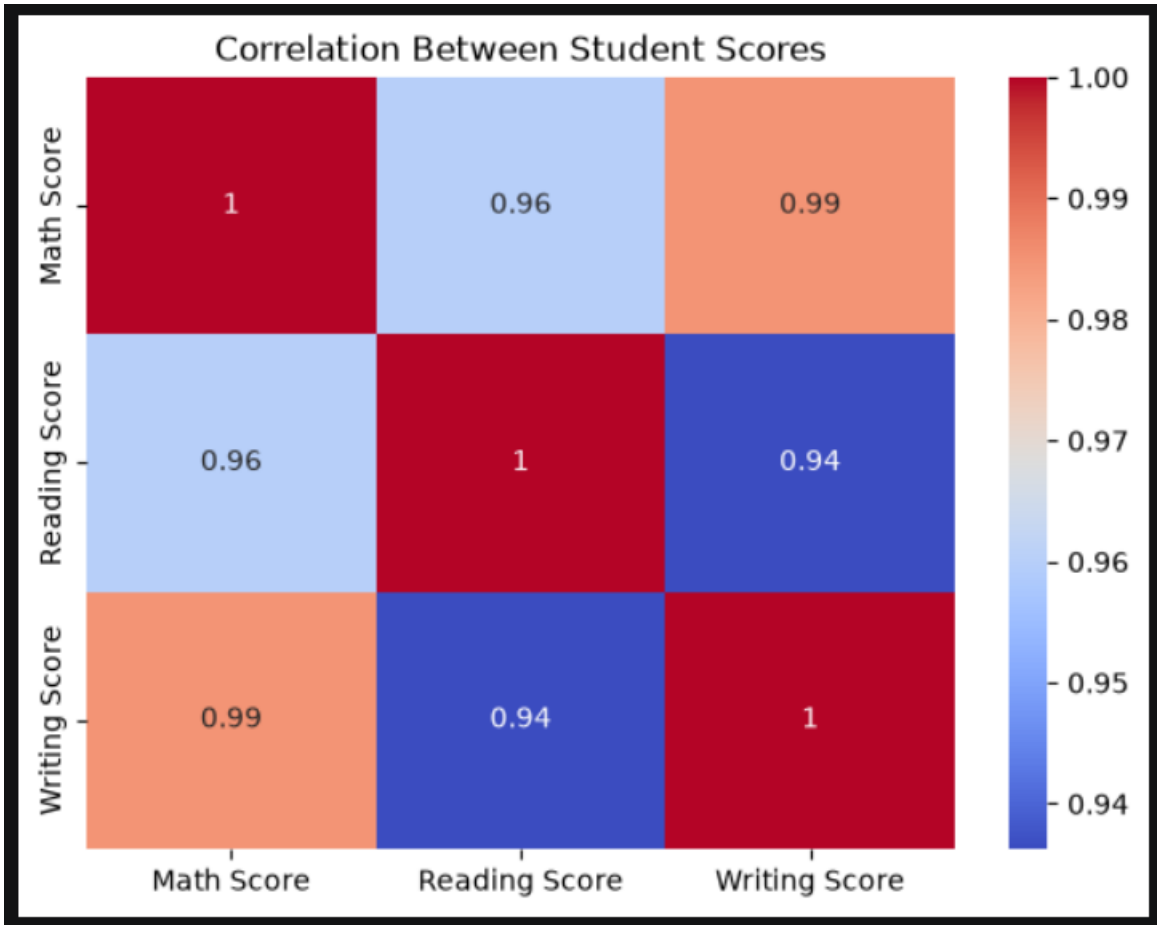
9. Tools and Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, and Jupyter Notebook.


Distribution of Math Scores
Number of Students
Math Score
0.0
0.5
1.0
1.5
2.0
2.5
3.0
60
65
70
75
80
85
90
95


Box Plot of Math Scores
Math Score
60
65
70
75
80
85
90
95


Math Score vs Reading Score
Reading Score
Math Score
90
85
80
75
70
65
60
65
70
75
80
85
90
95


Correlation Between Student Scores
Math Score
Reading Score
Writing Score
1
0.96
0.99
0.96
1
0.94
0.99
0.94
1
Math Score
Reading Score
Writing Score
1.00
0.99
0.98
0.97
0.96
0.95
0.94